

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Zoology

Paper: ZOB-601 — Mammalian Physiology

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. Draw diagrams wherever necessary. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False with proper justification:
[8 × 2 = 16]

- (i) The initial phase of the repolarization event in the action potential is achieved by opening of the voltage-gated potassium channel.
- (ii) The visual axis of the human eye crosses through optic disc during image formation.
- (iii) cGMP-gated Na channel in the cone cell is opened during scotopic vision.
- (iv) Almost 50% of the digestion of amylase takes place in buccal cavity.
- (v) All the absorption of lipids takes place in small intestine.
- (vi) Proteins are completely broken into amino acids before absorption.
- (vii) Transport of sodium ions across luminal membrane involves only antiport transport mechanism.
- (viii) ATPase action is associated with entire myosin filament.

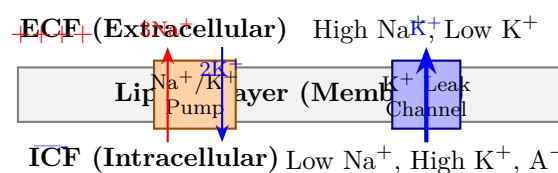
(b) Define the following:

[4 × 1 = 4]

- (i) Current sink
- (ii) Tonotopic map
- (iii) Regurgitation
- (iv) Anisotropic.

Question 2. Answer the following:

(a) What is membrane potential? Explain, how this is achieved using a suitable diagram. [7]

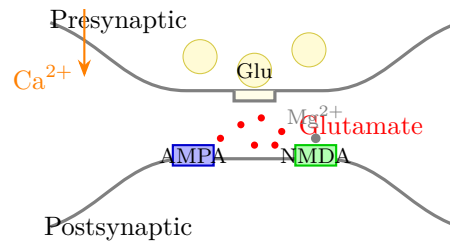


(b) Explain, how deficiency of vitamin B₁₂ may lead to Pernicious anaemia.

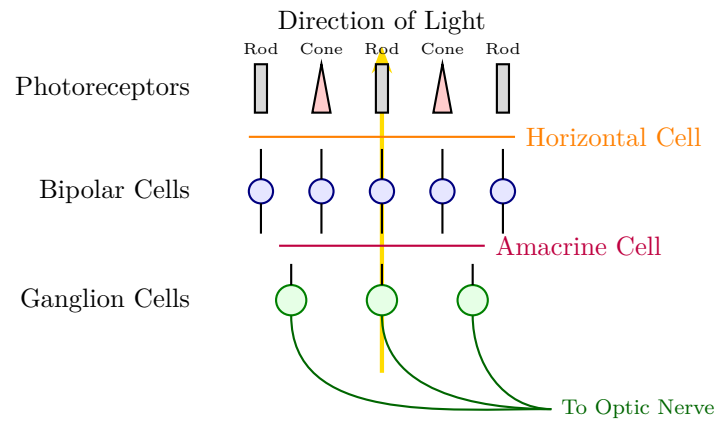
[3]

Question 3. Answer the following:

- (a) Using a systemic representation, enumerate the events occurring at the neuromuscular junction. Explain, how the above events are associated with Myasthenia Gravis. [6]
- (b) Draw a labeled sketch of a Glutamatergic synapse and briefly describe the mode of its functioning. [4]



Question 4. Draw a suitably labeled diagram of the human eye retina. Describe the basis of activation of rhodopsin and genesis of photoreceptor potential during dim light vision. [10]

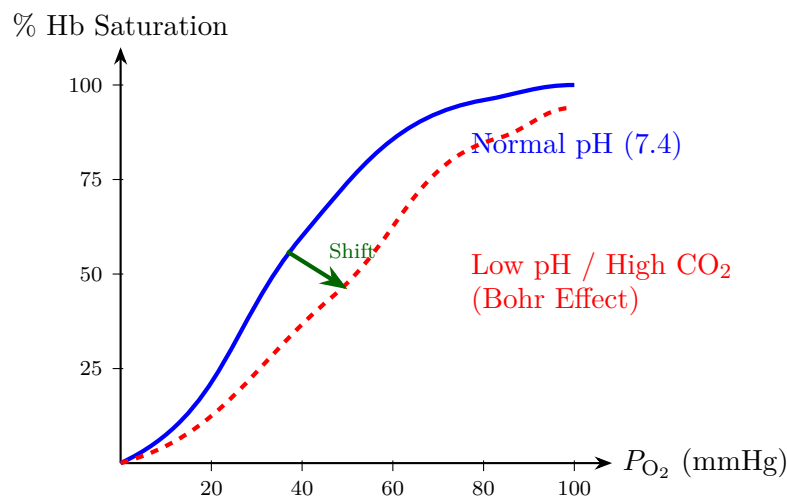


Question 5. Answer the following:

- (a) A person has met with an accident and has suffered heavy blood loss. Enumerate how the Renin-angiotensin-aldosterone system would respond to regulate blood pressure and fluid balance. [7]
- (b) What is impedance match? Briefly explain how this is achieved. [3]

Question 6. Answer the following:

- (a) What is the organ of Corti? Describe its structure and mechanism of function in hearing. [5]
- (b) What is Bohr's effect? Explain it by drawing a schematic graph representing the effect of CO_2 on the oxygen binding affinity of haemoglobin. How is blood pH related to this phenomenon? [5]



Question 7. Answer the following:

- (a) Arrange the respiratory quotients of proteins, carbohydrates and lipids in increasing order and explain the reason behind this difference. Can there be a metabolic situation where the respiratory quotient is more than 1.0? If yes, mention it. If no, explain why. [6]
- (b) Describe the cellular content of platelets. How do they contribute towards the formation of a platelet plug? [4]

Question 8. Answer the following:

- (a) Explain the complete process of digestion, absorption and assimilation of major dietary carbohydrates. [5]
- (b) Counter-current mechanism is the main adaptation for the concentration of urine. Explain the mechanism emphasizing on the anatomical features of the loop of Henle and the role of ADH. [5]

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